

VISION

Product: - CTCL

1.0.0

Introduction

Vision is an event-driven trading software designed to provide a complete ecosystem for market connectivity, order execution, risk management. This software has **no interface**. It runs **on terminal** and different logs tells different modules of software including logging, orderbook, order punch, RMS logs etc.

The software has different modules for different purposes

- Universe
- Market data connector
- Executor
- Order manager
- Risk manager
- Quote/Book manager
- Strategy
- Logger

Vision system config to load. This config tells how system will work and which streams to listen to and which executor to login etc..

```
1 {  
2   "timezone": 5.5,  
3   "logger": {  
4     "directory": "/logs/<date>",  
5     "over_write": false,  
6     "compressed_logging": false,  
7     "shared_memory_name": "logger_test",  
8     "size_in_MB": 16,  
9     "logger_name": "logger_test",  
10    "flush_buffer_size_in_KB": 32,  
11    "parallel_logger": true  
12  },  
13  "Hulk": {  
14    "realtime": true,  
15    "driver": "driver",  
16    "event_queue_polling_ratio": "<num>",  
17    "end_time": "<time>"  
18  },  
19  "base": {  
20    "portfolio": "/config/portfolio.json",  
21    "universe": "/config/universe.json",  
22    "mkt_data": "/config/connectors_prod.json"  
23  },  
24  "oms": "/config/oms.json",  
25  "predictors": "/config/predictors.json"  
26 }  
27
```

User Login into system

Only authorised users have access to the vision system.

Remote access denied

```
naveen@fd-vm024:~$ ssh dev2@192.168.25.101
dev2@192.168.25.101's password:
Permission denied, please try again.
dev2@192.168.25.101's password: █
```

access allowed

```
naveen@fd-vm024:~$ ssh dev2@192.168.25.101
dev2@192.168.25.101's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Fri 10 Oct 2025 12:57:35 PM IST

System load:  2.2               Processes:            391
Usage of /:   56.2% of 898.05GB Users logged in:      1
Memory usage: 5%               IPv4 address for enp1s0: 10.230.186.45
Swap usage:   0%               IPv4 address for enp1s0d1: 10.230.186.17
Temperature: 50.0 C           IPv4 address for enp5s0f0: 192.168.25.101

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Fri Oct 10 11:24:43 2025 from 192.168.10.50
. _ _ _ █
```

vision system access

```
dev2@apex:~/vision_v2/bin$ ll
total 80568
drwxrwxr-x  2 dev2 dev2    4096 Oct  9 21:31 ./
drwxr-xr-x 24 dev2 dev2    4096 Oct  9 21:51 ../
-rwxrwxr-x  1 dev2 dev2 82492064 Oct  9 21:31 vision_test*
dev2@apex:~/vision_v2/bin$ █
```

log in

```

config > {} executor.json > ...
1  {
2      "NSEFO": {
3          "ex1": {
4              "type": "nnf_nsefo",
5              "log": "ex1.log",
6              "hexdump_log": "ex1_hexdump.log",
7              "market_type": "TEST_MARKET",
8              "message_rate": 2,
9              "trader_id": "id",
10             "box_id": "id",
11             "branch_id": 1,
12             "PAN": "<pan>",
13             "version_number": "version",
14             "nnf_field": "nnf",
15             "algo_id": "id",
16             "user_id": "id",
17             "workstation_number": "",
18             "broker_id": "14697",
19             "password": "password",
20             "max_reconnects": 5,
21             "max_rejects": 10,
22             "local_ip_address": "<interface>",
23             "socket_type": "NONE",
24             "password_rotation_mode": false,
25             "download_from_dropcopy": false,
26             "run_as_dropcopy": false,
27             "encrypted_connection": true,
28             "certificate_filename": "config/FO_CA_Certificate/gr_cafo_cert5.pem"
29         }
30     }
31 }
32

```

This config ensures that only authenticate users are allowed to trade. Since Only authentic user have access to this confidential information like pan, box_id, user_id, password etc.

logs of login :-

```

logs > 20251008 > 20251008.ex1.log
1  0,154101.330534632,Executor,215,Set to broker mode
2  0,154101.330534632,sslConnectGateway,713,connecting to gateway server,IP address,172.19.245.107,Port,10267
3  0,154102.880538104,handleGatewayConnect,864,Connected to gateway router
4  0,154102.900536669,handleHandshake,665,SSL handshake is successful
5  0,154102.950537433,handleGatewayResponseRead,935,trans_code,2401,error,0,Broker ID,14697,IP address,172.19.245.145,Port,10250,Session Key,zDAM*3;X,
    Cryptographic Key,6PBi^Nc<M<P;San,H?^MMy>H>xlaFeT\,CryptographicIV,6u'tDEK-zzGNvPt@
6  0,154102.984009847,handleExchangeConnect,1011,connected to exchange
7  0,154103.298485828,parse,1492,num_streams,3
8  0,154103.298485828,parse,1521,minimum allowed disclosed percent is 100
9  0,154103.341212697,parse,1585,Download request sent,1,0
10 0,154103.341212697,parse,1585,Download request sent,2,0
11 0,154103.341212697,parse,1585,Download request sent,3,0
12 0,154103.396802804,parse,1860,Executor Ready
13

```

log out: -

when the software exit it log outs automatically.

The common Order lifecycle is

1. Order punch → order Check On RMS(vision system) → Order Acked → trade Acked/complete

Order Type can be of NEW or MODIFY.

This example is of test market in FO segment.

Order Punch :-

```
dev@apex:~/vision/logs$ tail -f 20250924.orders.csv
date,time,order_id,template_id,leg_id,exchange_order_id,action,status,error,user_tag,type,executor_id,exchange,symbol,side,price,open_size,visible_open_size,filled_size,last_trade_size,last_trade_price,last_trade_id,edate,etime,md_exchange,md_stream,md_sequence,md_index,transmit_date,transmit_time,trigger_event
0,155134,573118805,1,0,0,11000000000084880,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152235,179311925,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573173523,1,0,0,11000000000084880,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152736,147252650,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573205567,2,0,0,11000000000084856,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154134,958539687,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573250836,2,0,0,11000000000084856,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154500,010996376,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,618606112,3,0,0,0,OPEN,SENT,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,614729758,NSEFO,1,197384,0,0,155134,618608887,TOUCH
0,155134,652066911,3,0,0,1000000000078534,OPEN,ACKED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,634342448,NSEFO,1,197388,0,86397,051716.854775807,NONE
0,155134,652105115,3,0,0,1000000000078534,TRADE,COMPLETED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,0,0,10,10,10,18153,0,155134,634342448,NSEFO,1,197388,0,86397,051716.854775807,NONE
```

An order for ITC September expiry of strike 415.00 PUT option.

Order Acked:-

```
dev@apex:~/vision/logs$ tail -f 20250924.orders.csv
date,time,order_id,template_id,leg_id,exchange_order_id,action,status,error,user_tag,type,executor_id,exchange,symbol,side,price,open_size,visible_open_size,filled_size,last_trade_size,last_trade_price,last_trade_id,edate,etime,md_exchange,md_stream,md_sequence,md_index,transmit_date,transmit_time,trigger_event
0,155134,573118805,1,0,0,11000000000084880,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152235,179311925,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573173523,1,0,0,11000000000084880,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152736,147252650,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573205567,2,0,0,11000000000084856,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154134,958539687,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,573250836,2,0,0,11000000000084856,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154500,010996376,NSEFO,1,197383,0,86397,051716.854775807,NONE
0,155134,618606112,3,0,0,0,OPEN,SENT,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,614729758,NSEFO,1,197384,0,0,155134,618608887,TOUCH
0,155134,652066911,3,0,0,1000000000078534,OPEN,ACKED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,634342448,NSEFO,1,197388,0,86397,051716.854775807,NONE
0,155134,652105115,3,0,0,1000000000078534,TRADE,COMPLETED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,0,0,10,10,10,18153,0,155134,634342448,NSEFO,1,197388,0,86397,051716.854775807,NONE
```

Acked given by exchange.

Trade Complete: -

```
dev@apex:~/vision/logs$ tail -f 20250924.orders.csv
date,time,order_id,template_id,leg_id,exchange_order_id,action,status,error,user_tag,type,executor_id,exchange,symbol,side,price,open_size,visible_open_size,filled_size,last_trade_size,last_trade_price,last_trade_id,edate,etime,md_exchange,md_stream,md_sequence,md_index,transmit_date,transmit_time,trigger_event
0,155843,381176531,3,0,0,1000000000078534,OPEN,ACKED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,634342448,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,381238046,3,0,0,1000000000078534,TRADE,COMPLETED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,0,0,10,10,10,18153,0,155134,634342448,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408186341,1,0,0,11000000000084880,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152235,179311925,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408196911,1,0,0,11000000000084880,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152736,147252650,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408212684,2,0,0,11000000000084856,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154134,958539687,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408226366,2,0,0,11000000000084856,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154500,010996376,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408393376,4,0,0,0,OPEN,ERROR,RMS_LIMIT,0,0,NSEFO,WIPRO30SEP25,252,50CE,SELL,810,10,10,0,0,-9223372036854775808,0,0,155843,393330107,NSEFO,2,178935,0,-26355,053000,000000000,TOUCH
```

2. Order Punch → RMS check failed and Order was not punched

Order Punch:-

```
dev@apex:~/vision/logs$ cat 20250924.orders.csv
date,time,order_id,template_id,leg_id,exchange_order_id,action,status,error,user_tag,type,executor_id,exchange,symbol,side,price,open_size,visible_open_size,filled_size,last_trade_size,last_trade_price,last_trade_id,edate,etime,md_exchange,md_stream,md_sequence,md_index,transmit_date,transmit_time,trigger_event
0,155843,381176531,3,0,0,1000000000078534,OPEN,ACKED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,10,10,0,0,-9223372036854775808,0,0,155134,634342448,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,381238046,3,0,0,1000000000078534,TRADE,COMPLETED,NO_ERROR,0,0,NSEFO,ITC30SEP25,415,00PE,BUY,110,0,0,10,10,10,18153,0,155134,634342448,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408186341,1,0,0,11000000000084880,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152235,179311925,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408196911,1,0,0,11000000000084880,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,25600,00PE,BUY,38030,10,10,0,0,-9223372036854775808,0,0,152736,147252650,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408212684,2,0,0,11000000000084856,OPEN,ACKED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154134,958539687,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408226366,2,0,0,11000000000084856,CLOSE,COMPLETED,NO_ERROR,0,0,NSEFO,NIFTY30SEP25,24700,00CE,SELL,46000,10,10,0,0,-9223372036854775808,0,0,154500,010996376,NSEFO,1,208031,0,86397,051716.854775807,NONE
0,155843,408393376,4,0,0,0,OPEN,ERROR,RMS_LIMIT,0,0,NSEFO,WIPRO30SEP25,252,50CE,SELL,810,10,10,0,0,-9223372036854775808,0,0,155843,393330107,NSEFO,2,178935,0,-26355,053000,000000000,TOUCH
```

RMS Reject:-

```
dev@apex:~/vision/logs$ cat 20250924.rms.debug.log
0,155843,408393376,OrderId=4,order_action=OPEN,exch=NSEFO,ins=WIPRO30SEP25,252,50CE,side=SELL,newLots=0,origLots=10,order_price=8.100000,local_mode=NORMAL,exch_mode=NORMAL,inst_mode=NORMAL,inst_tradeable=1,inst_exchange_mode=NORMAL,checks, hits=TP_RANGE,
dev@apex:~/vision/logs$
```

Now let's walk through the different modules of Vision system.

Module: Universe (Instrument Loader and Event Generator)

This module is responsible for loading all tradable instruments into the system and managing event-based communication between components. During initialization, the

Universe reads configuration files and price data, constructs Instrument objects. Once loaded, the Universe logs instrument details and sets up data structures for efficient retrieval and market data updates.

The Event Manager manages event subscriptions, posting, and unsubscriptions while ensuring that listeners are executed in priority order. Whenever new market data or administrative actions occur, the Event Manager dispatches corresponding events to subscribed modules like strategies, risk managers, and execution handlers.

Config

```
config > {} universe.json > ...
1  {
2    "min_instruments": 0,
3    "max_instruments": 1000000000,
4    "enable_touch_logging": false,
5    "enable_trade_logging": false,
6    "group_pnl_by": "NONE",
7    "tickTableFile1": "/config/demotickTable.csv",
8    "pricesFile1": "/config/files/pricefile.csv",
9    "tickers": [
10     {
11       "universe_group": 0,
12       "exchange": "NSEFO",
13       "ha_model": "HA_HOLD",
14       "inst_types": ["FUTURES", "OPTION_CE", "OPTION_PE"],
15       "expiry_seq": [1,2,3,4],
16       "strike_num": 150,
17       "impact": true,
18       "quote_mkt_kind": "ORDER_BY_ORDER",
19       "tradeable": true
20     }
21   ]
22 }
23
```

Logs

1. Universe.csv: - this log shows how many instruments are loaded into vision system.

```
1 id,symbol,token,exchange,inst_type,underlying_type,underlying,underlying_exchange,expiry_date,expiry_seq,month_expiry_seq,strike_price,stream_index,lot_si
2 0,3600NE25NOV25_FUT,38313,NSEFO,FUTURES,EQUITY,3600NE,NSEFO,20251125,2,1,0,12,500,10,100,0,1072.200000,1,65535,NORMAL,-1
3 1,3600NE30DEC25_FUT,49545,NSEFO,FUTURES,EQUITY,3600NE,NSEFO,20251230,3,1,0,12,500,10,100,0,1071.400000,1,65535,NORMAL,-1
4 2,3600NE28OCT25_FUT,52170,NSEFO,FUTURES,EQUITY,3600NE,NSEFO,20251028,1,1,0,12,500,10,100,0,1068.800000,1,65535,NORMAL,-1
5 3,3600NE25NOV25_1340,00CE,35147,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251125,2,1,134000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
6 4,3600NE28OCT25_1520,00CE,35571,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,152000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
7 5,3600NE25NOV25_1400,00CE,35575,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251125,2,1,140000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
8 6,3600NE28OCT25_1560,00CE,37881,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,156000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
9 7,3600NE25NOV25_1480,00CE,44096,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251125,2,1,148000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
10 8,3600NE25NOV25_1480,00CE,44096,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251125,2,1,148000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
11 9,3600NE28OCT25_1480,00CE,46526,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,148000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
12 10,3600NE28OCT25_800,00CE,54260,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,80000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
13 11,3600NE28OCT25_740,00CE,54270,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,74000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
14 12,3600NE28OCT25_780,00CE,54272,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,78000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
15 13,3600NE28OCT25_820,00CE,54274,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,82000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
16 14,3600NE28OCT25_860,00CE,54276,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,86000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
17 15,3600NE28OCT25_900,00CE,54278,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,90000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
18 16,3600NE28OCT25_940,00CE,54280,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,94000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
19 17,3600NE28OCT25_980,00CE,54282,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,98000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
20 18,3600NE28OCT25_1020,00CE,54284,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,102000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
21 19,3600NE28OCT25_1060,00CE,54286,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,106000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
22 20,3600NE28OCT25_1100,00CE,54288,NSEFO,OPTION_CE,EQUITY,3600NE,NSEFO,20251028,1,1,110000,12,500,5,100,0,0.000000,1,65535,NORMAL,-1
```

Module: Market Data Connector

This module is responsible for connecting to multicast data feed, receiving market data packets in real-time, and parsing them into structured market events.

It manages multiple sessions per exchange stream, handling incremental, snapshot recovery and broadcast feed. The module interprets the binary data packets received from the exchange and applies them to corresponding instruments, updating their book, and raising different events.

Config

```
1 {
2   "stream1": {
3     "parser": {
4       "key": {
5         "type": "mtbt_nsefo"
6       }
7     },
8     "connector": {
9       "key": {
10        "type": "mtbt_nsefo",
11        "log": "stream1.csv",
12        "sessions": [
13          {
14            "market_type": "PRODUCTION",
15            "exchange": "NSEFO",
16            "ignore_begin_misses": false,
17            "stream_id": 1,
18            "mcast_local_address": "<interface>",
19            "recovery_local_address": "<interface>",
20            "recovery_socket_type": "NONE",
21            "use_snap_recovery": true,
22            "snap_dump_interval_in_minutes": "<num>",
23            "snap_recovery_min_missed_count": "<num>",
24            "msg_store_size_multiple_max": "<num>"
25          }
26        ]
27      }
28    }
29  }
30 }
31 }
```

Logs

1. Stream1.csv: -This log shows which streams we are connected to listen to the MTBT data.

```
logs > 20251008 > 20251008.stream1-2_test.csv > data
1 [Session::Session].Joining multicast for group,
2 [Session::Session].Joining multicast for group,
3 0,154102.861406035,requestRecovery,453,session,1,starting md recovery,start,0,end,79548
4 0,154102.861416970,requestRecovery,453,session,0,starting md recovery,start,0,end,91368
5 0,154102.866414241,startRecovery,630,session,1,session.recovery_sequence_number,79548,session.last_sequence_number,0,session.snap_recovery_min_missed_count,100
6 0,154102.866414241,startIncrementalRecovery,669,session,1,waiting for TCP connection,start sequence,0,end,79548
7 0,154102.866736330,startRecovery,630,session,0,session.recovery_sequence_number,91368,session.last_sequence_number,0,session.snap_recovery_min_missed_count,100
8 0,154102.866736330,startIncrementalRecovery,669,session,0,waiting for TCP connection,start sequence,0,end,91368
9 0,154102.890293473,handleRecoveryConnect,724,session,1,recovery connection established
10 0,154102.890293473,handleRecoveryConnect,724,session,1,requesting incremental recovery,start,1,end,79548
11 0,154102.890369946,handleRecoveryConnect,724,session,0,recovery connection established
12 0,154102.890369946,handleRecoveryConnect,745,session,0,requesting incremental recovery,start,1,end,91368
13 0,154102.914872659,handleRecoveryRequestResponse,776,session,0,recovery request accepted,start,0,end,91368
14 0,154102.915123313,handleRecoveryRequestResponse,776,session,1,recovery request accepted,start,0,end,79548
15 0,154105.871395187,handleErrorInRecovery,953,session,1,handleRecoveryRead,invoked with error=2 End of file
16 0,154105.871706873,stopRecovery,698,session,1,Recovery Done,1,Time :,0,154105.871733789
17 0,154105.871706873,stopRecovery,699,session,1,ending recovery,last sequence number,79601,session.recovery_sequence_number,79601
18 0,154105.980278679,stopRecovery,698,session,0,Recovery Done,0,Time :,0,154105.980589926
19 0,154105.980278679,stopRecovery,699,session,0,ending recovery,last sequence number,91702,session.recovery_sequence_number,91702
20
```

2. Orderbook: -

The top 5 level order book. This Order is of(NIFTY14OCT_25050 CE)

```
logs > {} 20251008.orderbook_NIFTY14OCT25_25050.CE
162 0,091500,0.028146452,etp=0,091500,0.028146452,SELL,PRICE:17705,INDEX=2,TOTALSIZE=1-[X_1400000000001416;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
163 0,091500,0.028146452,etp=0,091500,0.028146452,SELL,PRICE:17710,INDEX=3,TOTALSIZE=4-[X_1400000000001408;S_4;C_0;I_0;OP_-1]--{{{n0rd=1}}}
164 0,091500,0.028146452,etp=0,091500,0.028146452,SELL,PRICE:19980,INDEX=4,TOTALSIZE=1-[X_1400000000001780;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
165 0,091500,0.028146452,FINISH,Quote::raiseMktdata : Msg_type: 78,1591-----
166 0,091500,0.028151459,START,Quote::raiseMktdata : Msg_type: 78,1592-----
167 0,091500,0.028151459,etp=0,091500,0.028151459,BUY,PRICE:14660,INDEX=0,TOTALSIZE=1-[X_1400000000001631;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
168 0,091500,0.028151459,etp=0,091500,0.028151459,BUY,PRICE:14000,INDEX=1,TOTALSIZE=1-[X_1400000000001828;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
169 0,091500,0.028151459,etp=0,091500,0.028151459,BUY,PRICE:13215,INDEX=2,TOTALSIZE=1-[X_1400000000001699;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
170 0,091500,0.028151459,etp=0,091500,0.028151459,BUY,PRICE:12000,INDEX=3,TOTALSIZE=1-[X_1400000000001567;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
171 0,091500,0.028151459,etp=0,091500,0.028151459,BUY,PRICE:11175,INDEX=4,TOTALSIZE=4-[X_1400000000001955;S_2;C_0;I_0;OP_-1]-[X_1400000000001999;S_2;C_1;I_0;OP_-1]--{{{n0rd=2}}}
172 0,091500,0.028151459,etp=0,091500,0.028151459,SELL,PRICE:14945,INDEX=0,TOTALSIZE=1-[X_1400000000001954;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
173 0,091500,0.028151459,etp=0,091500,0.028151459,SELL,PRICE:15945,INDEX=1,TOTALSIZE=1-[X_1400000000001955;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
174 0,091500,0.028151459,etp=0,091500,0.028151459,SELL,PRICE:17705,INDEX=2,TOTALSIZE=1-[X_1400000000001416;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
175 0,091500,0.028151459,etp=0,091500,0.028151459,SELL,PRICE:17710,INDEX=3,TOTALSIZE=4-[X_1400000000001408;S_4;C_0;I_0;OP_-1]--{{{n0rd=1}}}
176 0,091500,0.028151459,etp=0,091500,0.028151459,SELL,PRICE:19980,INDEX=4,TOTALSIZE=1-[X_1400000000001780;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
177 0,091500,0.028151459,FINISH,Quote::raiseMktdata : Msg_type: 78,1592-----
178 0,091500,0.028154956,START,Quote::raiseMktdata : Msg_type: 78,1593-----
179 0,091500,0.028154956,etp=0,091500,0.028154956,BUY,PRICE:14660,INDEX=0,TOTALSIZE=1-[X_1400000000001631;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
180 0,091500,0.028154956,etp=0,091500,0.028154956,BUY,PRICE:14000,INDEX=1,TOTALSIZE=1-[X_1400000000001828;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
181 0,091500,0.028154956,etp=0,091500,0.028154956,BUY,PRICE:13215,INDEX=2,TOTALSIZE=1-[X_1400000000001699;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
182 0,091500,0.028154956,etp=0,091500,0.028154956,BUY,PRICE:12000,INDEX=3,TOTALSIZE=1-[X_1400000000001567;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
183 0,091500,0.028154956,etp=0,091500,0.028154956,BUY,PRICE:11175,INDEX=4,TOTALSIZE=4-[X_1400000000001955;S_2;C_0;I_0;OP_-1]-[X_1400000000001999;S_2;C_1;I_0;OP_-1]--{{{n0rd=2}}}
184 0,091500,0.028154956,etp=0,091500,0.028154956,SELL,PRICE:14945,INDEX=0,TOTALSIZE=1-[X_1400000000001954;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
185 0,091500,0.028154956,etp=0,091500,0.028154956,SELL,PRICE:15945,INDEX=1,TOTALSIZE=1-[X_1400000000001955;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
186 0,091500,0.028154956,etp=0,091500,0.028154956,SELL,PRICE:17705,INDEX=2,TOTALSIZE=1-[X_1400000000001416;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
187 0,091500,0.028154956,etp=0,091500,0.028154956,SELL,PRICE:17710,INDEX=3,TOTALSIZE=4-[X_1400000000001408;S_4;C_0;I_0;OP_-1]--{{{n0rd=1}}}
188 0,091500,0.028154956,etp=0,091500,0.028154956,SELL,PRICE:19980,INDEX=4,TOTALSIZE=1-[X_1400000000001780;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
189 0,091500,0.028154956,FINISH,Quote::raiseMktdata : Msg_type: 78,1593-----
190 0,091500,0.028163986,START,Quote::raiseMktdata : Msg_type: 78,1594-----
191 0,091500,0.028163986,etp=0,091500,0.028163986,BUY,PRICE:14660,INDEX=0,TOTALSIZE=1-[X_1400000000001631;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
192 0,091500,0.028163986,etp=0,091500,0.028163986,BUY,PRICE:14445,INDEX=1,TOTALSIZE=2-[X_1400000000002001;S_2;C_0;I_0;OP_-1]--{{{n0rd=1}}}
193 0,091500,0.028163986,etp=0,091500,0.028163986,BUY,PRICE:14000,INDEX=2,TOTALSIZE=1-[X_1400000000001828;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
194 0,091500,0.028163986,etp=0,091500,0.028163986,BUY,PRICE:13215,INDEX=3,TOTALSIZE=1-[X_1400000000001699;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
195 0,091500,0.028163986,etp=0,091500,0.028163986,BUY,PRICE:12000,INDEX=4,TOTALSIZE=1-[X_1400000000001567;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
196 0,091500,0.028163986,etp=0,091500,0.028163986,SELL,PRICE:14945,INDEX=0,TOTALSIZE=1-[X_1400000000001954;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
197 0,091500,0.028163986,etp=0,091500,0.028163986,SELL,PRICE:15945,INDEX=1,TOTALSIZE=1-[X_1400000000001955;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
198 0,091500,0.028163986,etp=0,091500,0.028163986,SELL,PRICE:17705,INDEX=2,TOTALSIZE=1-[X_1400000000001416;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
199 0,091500,0.028163986,etp=0,091500,0.028163986,SELL,PRICE:17710,INDEX=3,TOTALSIZE=4-[X_1400000000001408;S_4;C_0;I_0;OP_-1]--{{{n0rd=1}}}
200 0,091500,0.028163986,etp=0,091500,0.028163986,SELL,PRICE:19980,INDEX=4,TOTALSIZE=1-[X_1400000000001780;S_1;C_0;I_0;OP_-1]--{{{n0rd=1}}}
201 0,091500,0.028163986,FINISH,Quote::raiseMktdata : Msg_type: 78,1594-----
```

Module: Executor (Order Execution and Exchange Connectivity)

The Executor module is the core trading engine responsible for establishing and maintaining a live connection between the Vision system and the exchange gateways

Config

```
config > {} executor.json > ...
1  {
2      "NSEFO": {
3          "ex1": {
4              "type": "nnf_nsefo",
5              "log": "ex1.log",
6              "hexdump_log": "ex1_hexdump.log",
7              "market_type": "TEST_MARKET",
8              "message_rate": 2,
9              "trader_id": "id",
10             "box_id": "id",
11             "branch_id": 1,
12             "PAN": "<pan>",
13             "version_number": "version",
14             "nnf_field": "nnf",
15             "algo_id": "id",
16             "user_id": "id",
17             "workstation_number": "",
18             "broker_id": "14697",
19             "password": "password",
20             "max_reconnects": 5,
21             "max_rejects": 10,
22             "local_ip_address": "<interface>",
23             "socket_type": "NONE",
24             "password_rotation_mode": false,
25             "download_from_dropcopy": false,
26             "run_as_dropcopy": false,
27             "encrypted_connection": true,
28             "certificate_filename": "config/FO_CA_Certificate/gr_cafo_cert5.pem"
29         }
30     }
31 }
32
```

This config ensures that only authenticate users are allowed to trade. Since Only

authentic user have access to this confidential information.

logs

1. ex1.log: - this log shows all the activity related to executor from connecting to exchange and sending orders and receiving response.

```
logs > 20251008 > 20251008.ex1.log
1 0,154101.330534632,Executor,215,Set to broker mode
2 0,154101.330534632,sslConnectGateway,713,connecting to gateway server,IP address,172.19.245.107,Port,10267
3 0,154102.880538104,handleGatewayConnect,864,Connected to gateway router
4 0,154102.900536669,handleHandshake,665,SSL handshake is successful
5 0,154102.950537433,handleGatewayResponseRead,935,trans_code,2401,error,0,Broker ID,14697,IP address,172.19.245.145,Port,10250,Session Key,zDAM*3;X,
Cryptographic Key,6PBi~Nc<M<P;San,H?*MMy>H>xlaFeT\,CryptographicIV,6u'tDEK-zzGNvPt@
6 0,154102.984009847,handleExchangeConnect,1011,connected to exchange
7 0,154103.298485828,parse,1492,num_streams,3
8 0,154103.298485828,parse,1521,minimum allowed disclosed percent is 100
9 0,154103.341212697,parse,1585,Download request sent,1,0
10 0,154103.341212697,parse,1585,Download request sent,2,0
11 0,154103.341212697,parse,1585,Download request sent,3,0
12 0,154103.396002804,parse,1860,Executor Ready
13
```

2. hexdump.log: - this log shows packet send and receive in hexadecimal format of executor ex1.

```
logs > 20251009 > 20251009.ex1_hexdump.log
1 0,100608.514756956,SEND,2400,size: 70, data:
2 00: 00 46 00 00 00 01 e1 37 cf b1 ff f4 e9 a9 c5 35
3 10: 61 65 28 f9 5b 78 09 60 00 00 00 00 00 00 00
4 20: 85 56 00 00 00 00 00 00 00 00 00 00 00 00 00
5 30: 00 00 00 00 00 00 00 00 00 00 00 00 00 30 1a ee
6 40: 31 34 36 39 37 00
7
8 0,100608.548513958,RECV,2401,size: 146, data:
9 00: 00 92 00 00 00 00 69 50 7b 36 40 1c ec 2a f9 08
10 10: 27 e9 d1 4b 83 40 09 61 00 00 00 00 00 00 00
11 20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
12 30: 00 00 00 00 00 00 00 00 00 00 00 00 00 7c 1a ee
13 40: 31 34 36 39 37 00 31 37 32 2e 31 39 2e 32 34 35
14 50: 2e 31 34 37 00 00 00 00 28 0a 21 68 5a 70 4f 5f
15 60: 3b 4b 58 76 30 25 41 28 5b 29 5d 3b 5c 44 27 31
16 70: 48 77 66 48 4e 2c 76 6a 46 32 54 21 23 3e 7a 3e
17 80: 68 6d 4f 33 71 6f 54 4e 33 33 68 2b 56 2a 55 39
18 90: 3c 3c
19
20 0,100608.581340736,SEND,23008,size: 64, data:
21 00: 00 40 00 00 00 02 86 fa 6b 74 8a a8 1d bf 1f 0c
22 10: 18 16 4c 14 7d 9e 59 e0 00 00 00 00 00 00 00
23 20: 85 56 00 00 00 00 00 00 00 00 00 00 00 00 00
24 30: 00 00 00 00 00 00 00 00 00 00 00 00 00 2a 1a ee
25
```

Module: Order Manager (OMS)

The Order Manager is the central control unit for all trading operations within the Vision system. It manages order creation, modification, and cancellation workflows across exchanges while coordinating with executors and portfolio managers to ensure compliance and reliability.

Config

```
ontig > {} oms.json > ...
```

```
1 {
2   "executors": "/config/executor.json",
3   "enable_order_logging": true,
4   "cancel_one_by_one_interval": 100
5 }
6
```

Logs

1. Orders.csv: - this log shows all the order activity. This is order punch.

```
logs > 20251008 > 20251008.orders.csv > data
1 date,time,order_id,template_id,leg_id,exchange_order_id,action,status,error,user_tag,type,executor_id,exchange,symbol,side,price,open_size,visible_open_size,
175 0,154105.908650299,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,8,0,2,1,301500,4826,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
176 0,154105.908731025,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,7,0,3,1,301500,4827,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
177 0,154105.908797118,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,6,0,4,1,301500,4828,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
178 0,154105.908879488,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,5,0,5,1,301500,4829,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
179 0,154105.908949400,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,4,0,6,1,301500,4830,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
180 0,154105.909032538,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,3,0,7,1,301500,4831,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
181 0,154105.909094987,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,2,0,8,1,301500,4832,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
182 0,154105.909173844,1197,0,0,1100000000151628,TRADE,ACKED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,1,0,9,1,301500,4833,0,154105.877422415,NSEFO,1,89276,
0,86383,051716.854775807,NONE
183 0,154105.909244771,1197,0,0,1100000000151628,TRADE,COMPLETED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,BUY,301900,0,0,10,1,301500,4834,0,154105.877422415,NSEFO,1,
89276,0,86383,051716.854775807,NONE
184 0,154105.909316450,1192,0,0,1100000000151623,CLOSE,COMPLETED,NO_ERROR,0,GTC,0,NSEFO,TCS280CT25_FUT,SELL,302500,10,10,0,0,-9223372036854775808,0,0,154105.
877451580,NSEFO,1,89276,0,86383,051716.854775807,NONE
185
```

3. Execution.csv: - this log shows the trade activity of the vision system(our) orders.
Or This log is **trade book** in which our trades are logged.

```
logs > 20251008 > 20251008.execution.csv > data
1 date,time,order_id,exchange_order_id,user_id,exchange,symbol,token,side,price,size,lot_size,trade_id
73 0,154105.699926235,1196,1100000000151627,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4820
74 0,154105.699976020,1196,1100000000151627,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4821
75 0,154105.700058789,1196,1100000000151627,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4822
76 0,154105.700140936,1196,1100000000151627,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4823
77 0,154105.700232097,1196,1100000000151627,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4824
78 0,154105.908544528,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4825
79 0,154105.908650299,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4826
80 0,154105.908731025,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4827
81 0,154105.908797118,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4828
82 0,154105.908879488,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4829
83 0,154105.908949400,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4830
84 0,154105.909032538,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4831
85 0,154105.909094987,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4832
86 0,154105.909173844,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4833
87 0,154105.909244771,1197,1100000000151628,34134,NSEFO,TCS280CT25_FUT,52539,BUY,301500,1,175,4834
88
```

Module: Portfolio Manager (Risk, Limits, and Profit and Loss Supervision)

The **Portfolio Manager** module is the risk and margin control hub of the Vision system. It continuously tracks position-level and portfolio-level metrics — including turnover, exposure, margin utilization, realized/unrealized profit and loss, and order limits — to ensure all trading activities operate within configured risk thresholds.

User Risk Management Values

1. Order limits
 - Order min lots

- Order max lots
- Order max amounts
- Order last traded price range

2. Instrument wise limits

- Instrument max quantity
- Instrument max exposure
- Instrument min quantity
- Instrument max order size
- Instrument min exposure
- Instrument max turnover
- Instrument max gross open quantity
- Instrument max gross open amount
- Instrument max net open quantity
- Instrument max net open amount

3. Overall limits

- Max turnover
- Max net exposure
- Max gross exposure
- Max net open amount
- Max gross open amount
- Max margin
- Max orders

4. Kill Switch Limits

- Min profit and loss
- Max turnover change in a unit time
- Un-acked orders check in a unit time

Config

```

config > {} portfolio.json > end_time
1
2
3 "apply_margin_checks": true,
4 "use_global_portfolio": true,
5 "start_time": "<time>","end_time": "<time>",
6 "log_interval_in_seconds": 10,
7 "max_allowed_safety_log_lines": 5,
8 "rms_log_interval_in_seconds": 10,
9 "order_min_lots": 1,
10 "order_max_lots": 1000,
11 "order_max_amount": "10Cr",
12 "order_ltp_range": 1,
13 "inst_max_quantity": "10Cr",
14 "inst_max_exposure": "1000Cr",
15 "inst_min_quantity": "-10Cr",
16 "inst_max_order_size": "1Cr",
17 "inst_min_exposure": "-1000Cr",
18 "inst_max_turnover": "1000Cr",
19 "inst_max_gross_open_quantity": "10Cr",
20 "inst_max_gross_open_amount": "1000Cr",
21 "inst_max_net_open_quantity": "10Cr",
22 "inst_max_net_open_amount": "1000Cr",
23 "max_turnover": "10000Cr",
24 "max_net_exposure": "1000Cr",
25 "max_gross_exposure": "10000Cr",
26 "max_net_open_amount": "10Cr",
27 "max_gross_open_amount": "1000Cr",
28 "max_margin": "1000Cr",
29 "max_orders": "10Cr",
30 "killswitch": {
31   "ks_mode": "LIQUIDATE_ONLY",
32   "start_time": "<time>","end_time": "<time>",
33   "minpnl_interval_in_seconds": 5,
34   "unacked_interval_in_seconds": 1,
35   "turnover_change_interval_in_seconds": 1,
36   "max_margin_interval_in_seconds": 10,
37   "max_allowed_flutter_instance": 25,
38   "min_pnl": "-25Cr",
39   "max_turnover_change": "1000Cr",
40   "ks_cancel_one_by_one": false),
41 "check_invalid_book": false,
42 "exchange_cost_file": "/config/cost_file.csv",
43 "margin": { "NSEFO": {
44   "type": "inf_span_exposure_strategy",
45   "log_margin_details": true,
46   "log_margin_file": "marginfile.csv",
47   "add_premium_to_margin": true,
48   "span_file": "/config/files/NSEFO_span",
49   "exposure_percent_file": "config/files/exposurefile.csv",
50   "span_numbers": ["i01.csv","i02.csv","i03.csv","i04.csv"]
51 }
52 }

```

Logs

1. pnl.csv: - this log shows unrealised, realised, net profit and loss and other portfolio parameters like net exposure, net open amount, margin utilised.

```

logs > 20251008 > 20251008.rms_summary.csv > data
1 date,time,instrument,exchange,rms_check,count
2

```

2. Indpnl.csv: - this log shows instrument wise profit and loss and other portfolio parameters.

```

1 date,time,inst,exchange,ltp,net_pnl,realised,unrealised,cost,turnover,strike,turnover,quantity(lots),exposure,open_long_quantity,open_short_quantity,open_long_exposure,open_short_exposure,open_long_amount,
2 open_short_amount,market_turnover,mode,average_open_price,buy_turnover,sell_turnover
3 0.095228,838254080,NIFTY280CT25_FUT,NSEFO,25140.000000,-7662.179902,0.000000,0.000000,7662.179902,18855645.000000,18855645.000000,0.000000,1,0,0.000000,1885560.000000,0.000000,1885500.000000,297920640.000000,
4 NORMAL,25140.000000,9427822.500000,9427822.500000
5 0.095228,838254080,NIFTY280CT25_24750.00CE,NSEFO,478.900000,-243.131228,0.000000,-191.250000,51.881228,72026.250000,72026.250000,2,-71835.000000,1,1,35917.500000,35917.500000,35823.750000,35655.000000,72026.
6 250000,NORMAL,480.175000,72026.250000,0.000000
7 0.095238,838254944,NIFTY280CT25_FUT,NSEFO,25140.100000,-9825.746099,52.500000,9660.746099,24512122.500000,24512122.500000,1,-1885507.500000,1,1,1885507.500000,1885507.500000,1885492.500000,1885350.
8 000000,333744630.000000,NORMAL,25139.000000,13198747.500000,11313375.000000
9 0.095238,838254944,NIFTY280CT25_24750.00CE,NSEFO,478.900000,-243.131228,0.000000,-191.250000,51.881228,72026.250000,72026.250000,2,-71835.000000,1,1,35917.500000,35917.500000,35756.250000,35658.750000,72026.
10 250000,NORMAL,480.175000,72026.250000,0.000000
11 0.095248,838253998,NIFTY280CT25_FUT,NSEFO,25138.100000,-19876.925101,753.750000,56.250000,20686.925101,50907877.500000,50907877.500000,-1,1885357.500000,0,1,1885357.500000,0.000000,1885417.500000,0.000000,
12 494005117.500000,NORMAL,25138.050000,24510855.000000,26397022.500000
13 0.095248,838253998,NIFTY280CT25_24750.00CE,NSEFO,478.900000,-243.131228,0.000000,-191.250000,51.881228,72026.250000,72026.250000,2,-71835.000000,1,1,35917.500000,35917.500000,35718.750000,35688.750000,72026.
14 250000,NORMAL,480.175000,72026.250000,0.000000
15 0.095258,838254802,NIFTY280CT25_FUT,NSEFO,25139.200000,-28711.500883,1386.000000,549.000000,30646.500883,75417120.000000,75417120.000000,4,-7541760.000000,0,1,1885440.000000,0.000000,1885642.500000,0.000000,
16 699513420.000000,NORMAL,25137.370000,41478472.500000,33938647.500000
17 0.095258,838254802,NIFTY280CT25_24750.00CE,NSEFO,478.000000,-611.031039,-457.500000,0.000000,153.531039,143595.000000,3856095.000000,0,0.000000,1,1,35850.000000,35850.000000,36067.500000,35800.000000,143595.
18 000000,NORMAL,478.000000,72026.250000,71568.750000
19 0.095308,838255060,NIFTY280CT25_FUT,NSEFO,25135.000000,-32734.808026,1386.000000,-1176.000000,32944.008026,81072960.000000,81072960.000000,7,-13195875.000000,0,1,1885125.000000,0.000000,1885350.000000,0.000000,
20 735332610.000000,NORMAL,25137.240000,47134312.500000,33938647.500000
21 0.095308,838255060,NIFTY280CT25_24750.00CE,NSEFO,478.400000,-636.875762,-457.500000,0.000000,179.375762,179475.000000,3891975.000000,1,-35880.000000,1,1,35880.000000,35880.000000,35782.500000,35625.000000,

```


3. Marginfile.csv: - this log shows margin utilised underlying wise and overall wise.

```
logs > 20251008 > 20251008.marginfile.csv > data
```

1	date	time	name	expiry	span	margin	exposure	margin	net	premium	total	margin
74801	0	,154105.909244771	,RVNL	-	,0.000000	,0.000000	,0.000000	,0.000000				
74802	0	,154105.909244771	,KFINTTECH	-	,0.000000	,0.000000	,0.000000	,0.000000				
74803	0	,154105.909244771	,ITC	-	,0.000000	,0.000000	,0.000000	,0.000000				
74804	0	,154105.909244771	,360ONE	-	,0.000000	,0.000000	,0.000000	,0.000000				
74805	0	,154105.909244771	,JUBLFOOD	-	,0.000000	,0.000000	,0.000000	,0.000000				
74806	0	,154105.909244771	,IRFC	-	,0.000000	,0.000000	,0.000000	,0.000000				
74807	0	,154105.909244771	,KAYNES	-	,0.000000	,0.000000	,0.000000	,0.000000				
74808	0	,154105.909244771	,SIEMENS	-	,0.000000	,0.000000	,0.000000	,0.000000				
74809	0	,154105.909244771	,KPITTECH	-	,0.000000	,0.000000	,0.000000	,0.000000				
74810	0	,154105.909244771	,PIDILITIND	-	,0.000000	,0.000000	,0.000000	,0.000000				
74811	0	,154105.909244771	,CUMMINSIND	-	,0.000000	,0.000000	,0.000000	,0.000000				
74812	0	,154105.909244771	,LAURUSLABS	-	,0.000000	,0.000000	,0.000000	,0.000000				
74813	0	,154105.909244771	,JINDALSTEL	-	,0.000000	,0.000000	,0.000000	,0.000000				
74814	0	,154105.909244771	,LICHSGFIN	-	,0.000000	,0.000000	,0.000000	,0.000000				
74815	0	,154105.909244771	,KOTAKBANK	-	,0.000000	,0.000000	,0.000000	,0.000000				
74816	0	,154105.909244771	,GODREJCP	-	,0.000000	,0.000000	,0.000000	,0.000000				
74817	0	,154105.909244771	,LT	-	,0.000000	,0.000000	,0.000000	,0.000000				
74818	0	,154105.909244771	,IGL	-	,0.000000	,0.000000	,0.000000	,0.000000				
74819	0	,154105.909244771	,LTIM	-	,0.000000	,0.000000	,0.000000	,0.000000				
74820	0	,154105.909244771	,SHRIRAMFIN	-	,0.000000	,0.000000	,0.000000	,0.000000				
74821	0	,154105.909244771	,MANAPPURAM	-	,0.000000	,0.000000	,0.000000	,0.000000				
74822	0	,154105.909244771	,OVERALL	-	,5246797.500000	,1250136.825000	,0.000000	,6496934.325000				
74823	0	,154111.330534859	,OVERALL	-	,5246797.500000	,1250136.825000	,0.000000	,6496934.325000				
74824	0	,154121.330534914	,OVERALL	-	,5246797.500000	,1250136.825000	,0.000000	,6496934.325000				
74825	0	,154122.460534810	,OVERALL	-	,5246797.500000	,1250136.825000	,0.000000	,6496934.325000				
74826												

Module: Book Manager

The **Quote/Book Manager** forms the market data core of the trading engine, responsible for constructing and maintaining a real-time view of the order book for every instrument.

logs

1. touch.csv: - this log shows market best bid, best ask changes.

```
data2 > logs > test_market > 20251003 > 20251003.touch.csv > data
```

1	local_date	local_time	exchange_date	exchange_time	symbol	inst	exchange	best_bid	best_ask	best_bid_size	best_ask_size	exchange	stream_id	sequence
2	0	,121111.392082892	,0	,100301.890777704	,BANKNIFTY28OCT25_FUT	,NSEFO	,5200000	,9223372036854775808	,1	,0	,NSEFO	,2	,1	
3	0	,121111.392134078	,0	,100301.922156589	,BANKNIFTY28OCT25_FUT	,NSEFO	,5300000	,9223372036854775808	,1	,0	,NSEFO	,2	,2	
4	0	,121111.392148925	,0	,100301.953378030	,BANKNIFTY28OCT25_FUT	,NSEFO	,5300000	,5900000	,1	,1	,NSEFO	,2	,3	
5	0	,121111.392163090	,0	,100301.985142570	,BANKNIFTY28OCT25_FUT	,NSEFO	,5300000	,5700000	,1	,1	,NSEFO	,2	,4	
6	0	,121111.392175971	,0	,100751.337683480	,NIFTY28OCT25_24000.00CE	,NSEFO	,96905	,9223372036854775808	,1	,0	,NSEFO	,2	,5	
7	0	,121111.392190347	,0	,100752.350020653	,NIFTY28OCT25_24000.00CE	,NSEFO	,96910	,9223372036854775808	,1	,0	,NSEFO	,2	,6	
8	0	,121111.392201996	,0	,100759.552130194	,NIFTY28OCT25_24000.00CE	,NSEFO	,9223372036854775808	,9223372036854775808	,0	,0	,NSEFO	,2	,7	
9	0	,121111.392210728	,0	,100801.154506396	,NIFTY28OCT25_24000.00CE	,NSEFO	,96710	,9223372036854775808	,1	,0	,NSEFO	,2	,8	
10	0	,121111.392218729	,0	,100815.580044437	,NIFTY28OCT25_24000.00CE	,NSEFO	,96760	,9223372036854775808	,1	,0	,NSEFO	,2	,9	
11	0	,121111.392227458	,0	,100846.635662200	,NIFTY28OCT25_24000.00CE	,NSEFO	,96860	,9223372036854775808	,1	,0	,NSEFO	,2	,10	
12	0	,121111.392236010	,0	,100848.239003273	,NIFTY28OCT25_24000.00CE	,NSEFO	,96905	,9223372036854775808	,1	,0	,NSEFO	,2	,11	
13	0	,121111.392249721	,0	,100923.042537274	,NIFTY28OCT25_24000.00CE	,NSEFO	,9223372036854775808	,9223372036854775808	,0	,0	,NSEFO	,2	,13	
14	0	,121111.392262718	,0	,100923.090021489	,NIFTY28OCT25_24100.00CE	,NSEFO	,9223372036854775808	,86700	,0	,1	,NSEFO	,2	,14	
15	0	,121111.392272111	,0	,100923.288119510	,NIFTY28OCT25_24200.00CE	,NSEFO	,9223372036854775808	,77820	,0	,1	,NSEFO	,2	,15	
16	0	,121111.392285531	,0	,100924.312026659	,NIFTY28OCT25_24200.00CE	,NSEFO	,9223372036854775808	,77815	,0	,6	,NSEFO	,2	,16	
17	0	,121111.392294478	,0	,100924.349960875	,NIFTY28OCT25_24200.00CE	,NSEFO	,9223372036854775808	,77810	,0	,6	,NSEFO	,2	,17	
18	0	,121111.392303010	,0	,100924.432447776	,NIFTY28OCT25_24100.00CE	,NSEFO	,9223372036854775808	,86695	,0	,1	,NSEFO	,2	,18	
19	0	,121111.392311501	,0	,100925.035000570	,NIFTY28OCT25_24100.00CE	,NSEFO	,9223372036854775808	,77805	,0	,6	,NSEFO	,2	,19	

2. trade.csv: - this log shows market trade messages.


```

1 local_date,local_time,exchange_date,exchange_time,symbol,inst_exchange,trade_price,trade_size,boid,soid,agg_side,flag,exchange,stream_id,sequence
2 0,121111.39,Col 2: local_time,042537274,NIFTY28OCT25_24000.00CE,NSEFO,96905,1,1100000000000029,1100000000000030,AGG_SELL,N,NSEFO,2,13
3 0,121111.39,250047166,NIFTY28OCT25_24200.00CE,NSEFO,77670,6,1100000000000033,1100000000000032,AGG_BUY,N,NSEFO,2,70
4 0,121111.392920637,0,101007.221134231,NIFTY28OCT25_24100.00CE,NSEFO,86515,1,1100000000000087,1100000000000031,AGG_BUY,N,NSEFO,2,90
5 0,121111.392967842,0,101244.237746908,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000155,1100000000000156,AGG_SELL,N,NSEFO,2,95
6 0,121111.392992181,0,101349.547055332,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000000157,1100000000000158,AGG_SELL,N,NSEFO,2,98
7 0,121111.393082531,0,102158.877302226,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000293,1100000000000294,AGG_SELL,N,NSEFO,2,98
8 0,121111.393105429,0,102214.557002073,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000000295,1100000000000296,AGG_SELL,N,NSEFO,2,107
9 0,121111.393128712,0,102253.322918727,BANKNIFTY28OCT25_FUT,NSEFO,5566200,1,1100000000000297,1100000000000298,AGG_SELL,N,NSEFO,2,113
10 0,121111.393505081,0,104453.398829693,TATASTEEL28OCT25_FUT,NSEFO,16800,1,1100000000001572,1100000000003524,AGG_SELL,N,NSEFO,2,153
11 0,121111.393607177,0,104620.701636040,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,163
12 0,121111.393615764,0,104620.701636040,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,164
13 0,121111.393623318,0,104620.801276938,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,165
14 0,121111.393630910,0,104620.801276938,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,166
15 0,121111.393646091,0,104620.901351324,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,167
16 0,121111.393654601,0,104620.901351324,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,168
17 0,121111.393662205,0,104621.001317720,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,169
18 0,121111.393669689,0,104621.001317720,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,170
19 0,121111.393677156,0,104621.101449209,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,171
20 0,121111.393684405,0,104621.101449209,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,172
21 0,121111.393691600,0,104621.201486800,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,173
22 0,121111.393698890,0,104621.201486800,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,174
23 0,121111.393719005,0,104621.301397869,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,176
24 0,121111.393727099,0,104621.301397869,BANKNIFTY28OCT25_54500.00PE,NSEFO,60000,1,1100000000000299,0,AGG_SELL,N,NSEFO,2,177
25 0,121111.393734581,0,104621.401663880,BANKNIFTY28OCT25_54500.00CE,NSEFO,60000,1,1100000000003514,0,AGG_SELL,N,NSEFO,2,178

```

Module: Strategy (Event-Driven Decision Engine)

The **Strategy module** serves as the intelligence layer of the Vision system, responsible for decision-making and automated trade execution. It operates by subscribing to real-time market and system events through the Event Manager. Whenever a new tick, trade, or signal event is received, the strategy processes it using predefined logic — performing computations — and then triggers corresponding actions like placing, modifying, or cancelling orders through the Order Manager. Each strategy runs in an isolated context, maintaining its own parameters, position state, and execution history, while sharing access to common modules like the Portfolio Manager and Universe for data and risk validation

every strategy can have different config and log.

Module: Logger

This module in the vision system is designed to operate as an independent, that efficiently manages all logging activities without interfering with the main program's execution. The Logger continuously monitors a shared queue or buffer where other modules push their log messages. It processes these entries in sequence, formatting and writing them to output destinations such as files or CSV logs, depending on configuration.

All the logs created by vision system are transferred into NAS cloud storage and are archived.